STAT 636 · FINAL PROJECT · GROUP 17

Centrality-Based Epidemic Control in a Hospital Ward Contact Network

Who should a ward isolate first when it can isolate only a few people?

Arvinder Singh Mundra · Hasini Kavuru · Satvika Reddy Akati
Texas A&M University · Fall 2026

INTRODUCTION

Isolation capacity is scarce. Who should go first?

The busiest people

Many contacts, or long hours of contact. The usual instinct is to isolate them first.

Degree · eigenvector centrality

The bridges

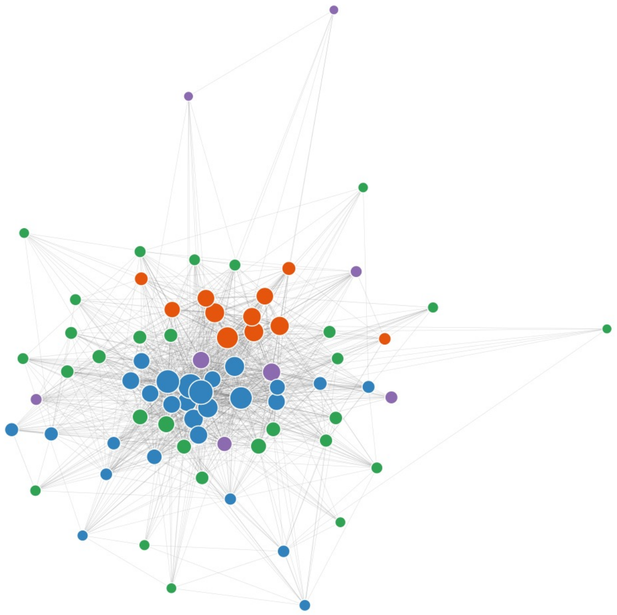
Fewer contacts, but they sit on short routes between people or groups that rarely meet.

Betweenness · harmonic centrality

Which centrality-based isolation rule most reduces the expected outbreak size, and does it still work when contact information is incomplete?

INTRODUCTION · DATA

Four days of sensor contacts in a geriatric ward



75
people

32,424
20-second contacts

1,139
edges (density 0.41)

- Lyon, Dec 6–10, 2010: 29 patients, 27 nurses and aides, 11 doctors, 8 administrators
- Edge weight = contact hours per day (total hours ÷ 4.02 days)
- Dense: mean degree 30, diameter 3; median person 0.6 h/day, busiest nurse 5.9

75%
of patients' contact time is with nurses

5%
is with other patients

75%
of doctors' contact time is with doctors

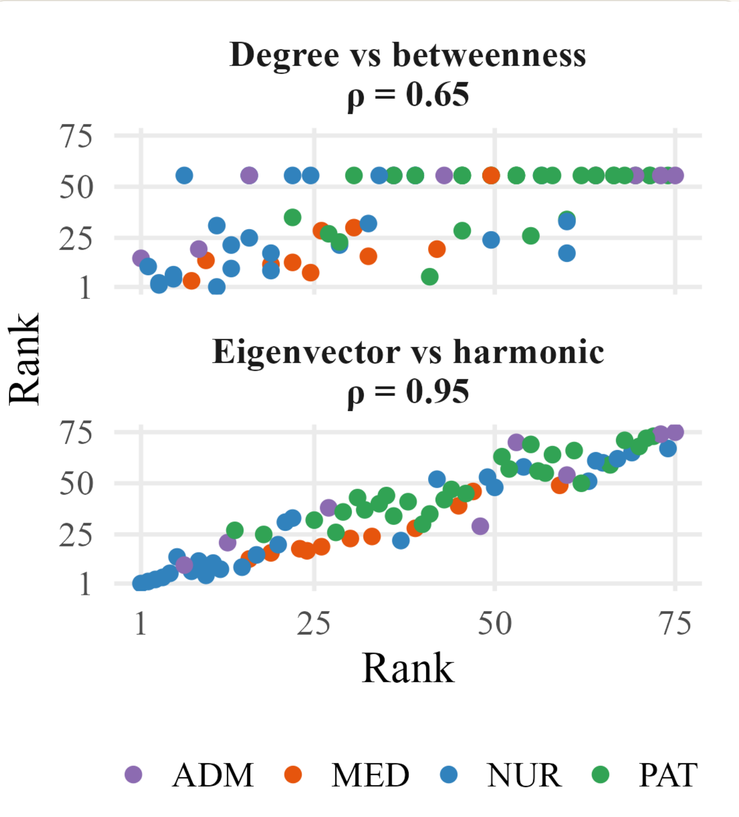
Data: SocioPatterns hospital ward contact network (Vanhems et al., 2013, PLOS ONE), CC BY-NC-SA · 3

METHODS

Four centrality measures, four different rankings

Measure	What it rewards	Ranked #1
Degree	Number of distinct contacts	An administrator
Eigenvector	Contact hours with well-connected people	A nurse
Betweenness	Lying on shortest paths (length = 1 / hours)	A nurse
Harmonic	Being close to everyone (length = 1 / hours)	A nurse

Degree's top pick has 61 contacts but only 2.0 contact hours a day. Each rule isolates its top k, ranked once, ties broken by a fixed random order. Baseline: k people at random, redrawn every run.



Spearman ρ across all pairs: 0.59 to 0.95

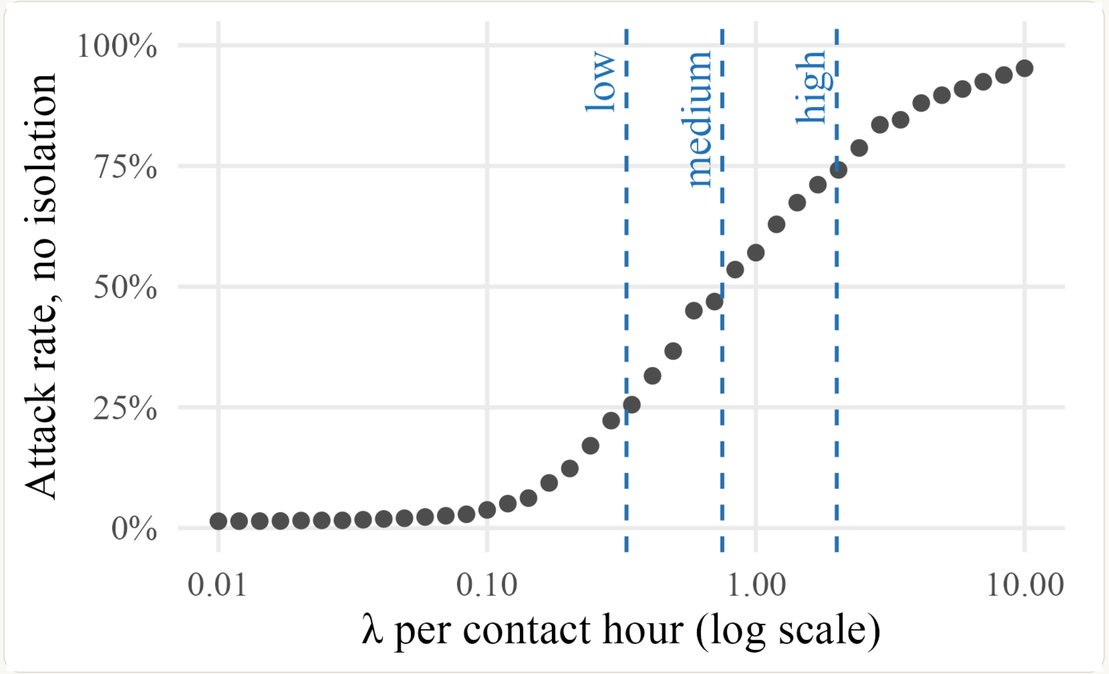
METHODS

A weighted SIR model on daily contact hours

$$P(j \text{ infected today}) = 1 - \exp(-\lambda \cdot \Sigma_i \, w_{ij})$$

sum over people i infected at the start of the day; w = daily contact hours between i and j

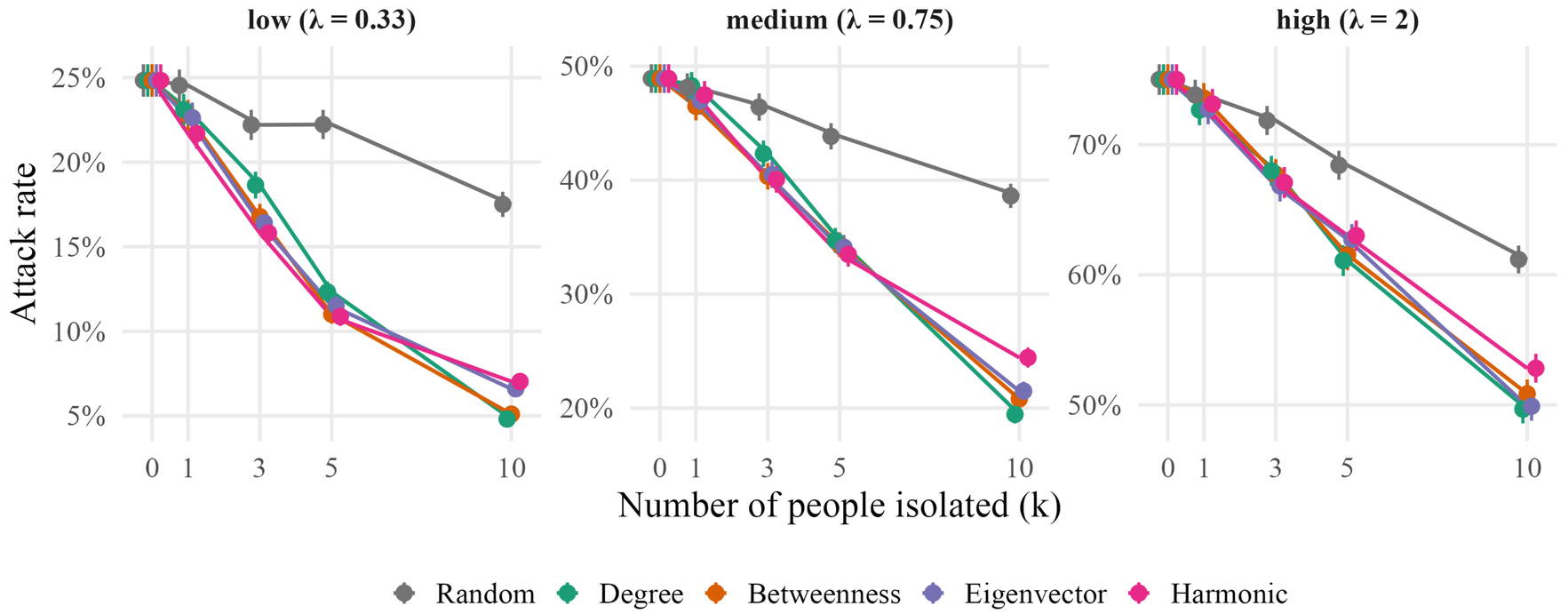
- One step = one day; recovery probability 0.2 (mean 5 infectious days)
- λ fixed from no-isolation pilots before any comparison: 0.33, 0.75, 2.0 → about 25%, 50%, 75% infected
- Isolate k = 1, 3, 5, 10; first case drawn from those not isolated
- 3,000 runs per condition (proposal: 500); simulator passed 8 exact checks



Outcome: attack rate = everyone ever infected ÷ 75 · secondary: P(at least 15 infected) · 95% Monte Carlo intervals · 5

RESULTS

Every centrality rule beats random isolation

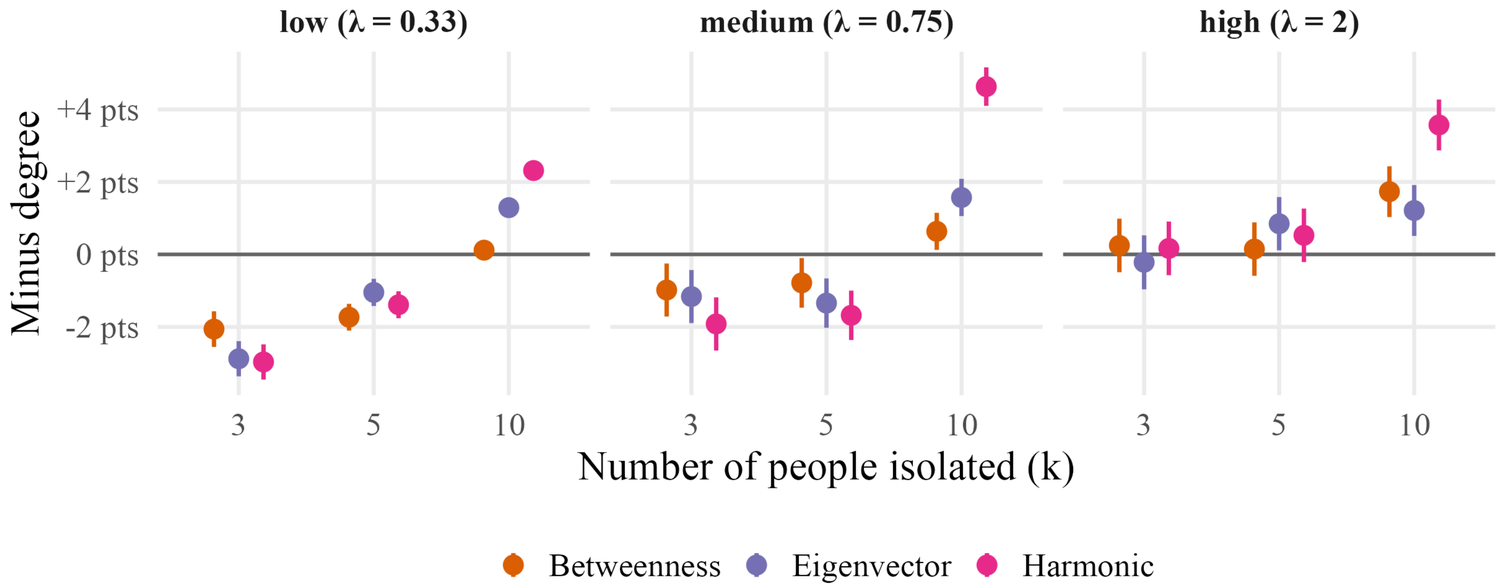


Isolating 10 people: **4.8–7.0%** infected vs 17.5% at random (low) · **19.4–24.4%** vs 38.6% (medium) · **49.7–52.8%** vs 61.2% (high). One person: at most 3.1 points.

3,000 runs per point · bars = 95% Monte Carlo intervals · 6

RESULTS

The best rule depends on how many can be isolated



Isolate 3–5 people

Weighted measures beat degree by 1–3 points at low and medium transmission

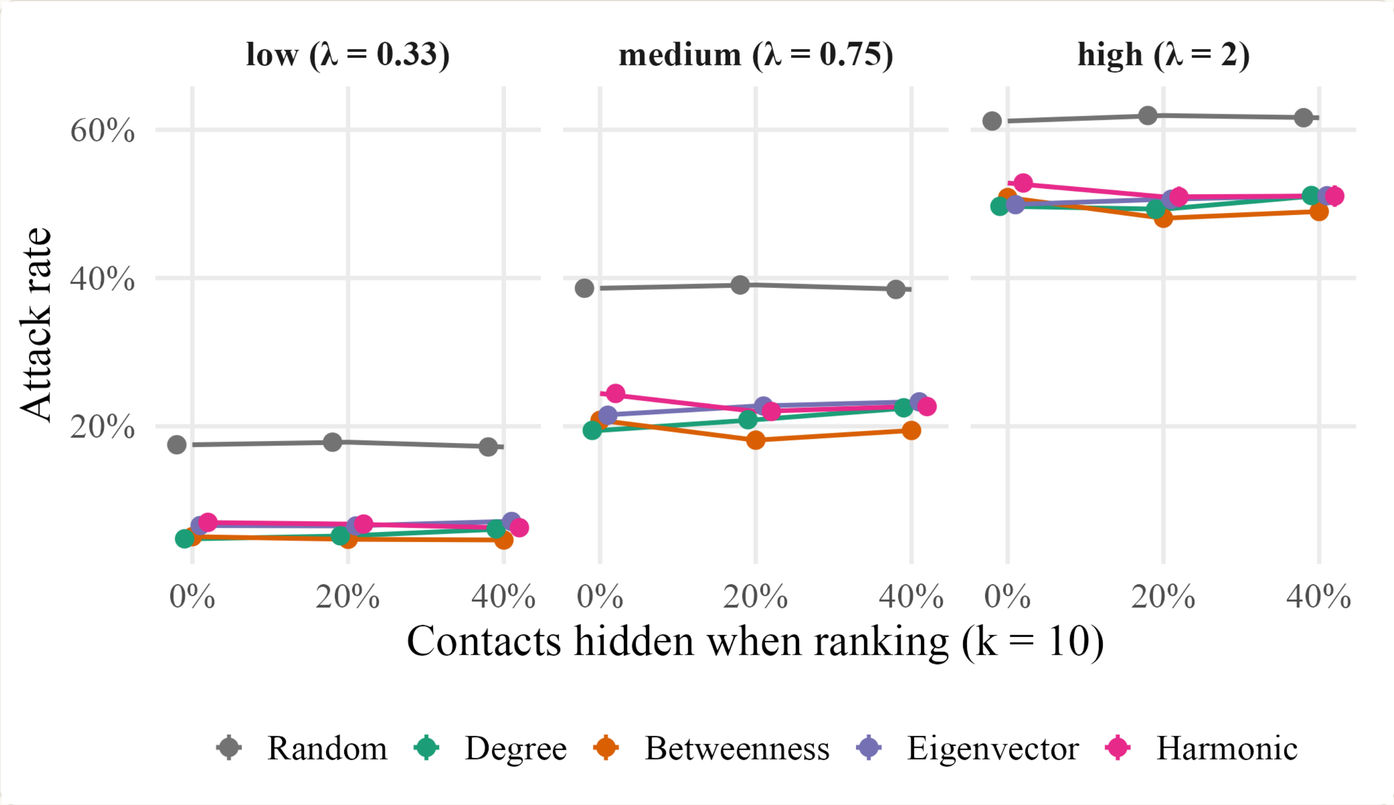
Isolate 10 people

Degree is lowest at every level (betweenness close); harmonic is 2–5 points worse

5 independent replicates × 3,000 runs; patterns held in at least 4 of 5 replicates · 7

RESULTS

Hiding up to 40% of contacts barely hurts



Rank on a network with 20% or 40% of edges hidden at random; simulate spread on the full network. 30 shared samples × 100 runs.

30–92%

overlap with the complete-data choice: **who** is picked changes a lot

±3 points

change in attack rate: **how well** it works hardly changes

Intervals use the spread across the 30 missing-edge samples (epidemic + missing-data variation) · 8

RESULTS

Interpreting the pattern: communities and rewiring

3 spectral communities

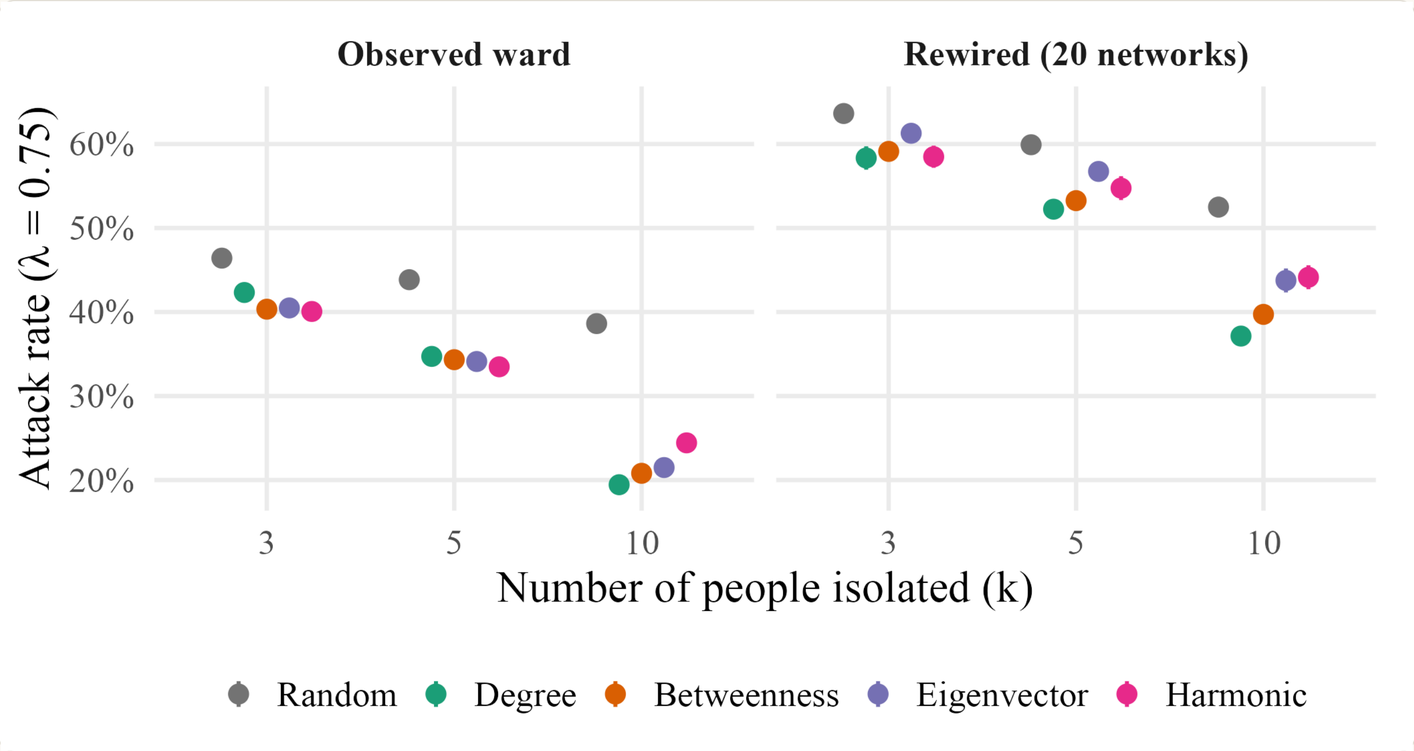
Doctors with their patients, a large nurse group, a small nurse–patient group. They mix roles (ARI 0.17–0.22).

Concentrated picks

All top-10 eigenvector and harmonic picks are in the nurse group; degree and betweenness reach two communities.

Rewired check

Same degrees, communities gone: outbreaks grow and degree is best or tied at every *k*.



Rewiring: 20 degree-preserving networks, weights permuted at random, medium transmission (λ = 0.75) · 9

DISCUSSION

Open questions, and how to address them

Pick sets, not people
Greedy rule: add the person whose isolation most lowers the simulated attack rate, given those already chosen.

Contact timing
Run a temporal SIR on the time-stamped records and compare the ranking of rules.

Realistic missing data
Hide the shortest contacts first instead of random ones.

Limits
One ward, four days, illustrative model, no replacement staff.

TAKEAWAYS

1 · Any centrality rule beats random isolation: 37–73% fewer infections with 10 isolated (low and medium).

2 · Few slots: use contact-time measures. Ten slots: plain degree does as well or better.

3 · Missing contacts change who is chosen, but hardly how well isolation works.